Query · normal → 90m / 6h / 10h

One question is a vector in $\mathbb{R}^{768}$. Each time $t \in \{1, 2, 3\}$ draws its own 128 gene questions, then appends 1 cell question.

1 Which genes

96 uniform from genes in x^{src}

32 from genes absent in x^{src}

$w(g) = P(g \text{ on} \mid \text{type}, t)^{1/2} + 0.02$

this cell's own target list is not used

a fresh draw at $t = 1$, at $t = 2$, and at $t = 3$

2 Source rank

$r_{\text{src}}(g)$ = index of g in x^{src}

$R[r] \in \mathbb{R}^{768}$, learned

table $R \in \mathbb{R}^{2049 \times 768}$

rows $0, \ldots, L_s - 1$ are real ranks, $L_s \le 512$

$r = -1$ uses the last row, one shared code

3 Target time

$T[t] \in \mathbb{R}^{768}$, learned

table $T \in \mathbb{R}^{4 \times 768}$

rows 1, 2, 3 are 90m, 6h, 10h

the same vector is added to every question

the encoder's time index stays fixed at 1

$$q = E[g] + R[r_{\text{src}}(g)] + T[t] \in \mathbb{R}^{768}$$

$E[g] \in \mathbb{R}^{768}$ is g's row in the encoder token table: the gene's name. Context stays in H_{src}, which the predictor reads as memory.

In the source

77 of the 96

$E[g]$	token row
$R[r]$	its own rank
$T[t]$	this time

Mask-CE

19 of the 96, token → ⟨mask⟩

m	learned, no name
$R[r]$	rank stays
$T[t]$	time stays

Absent from source

the 32

$E[g]$	name stays
$R_{\varnothing}$	the last row
$T[t]$	this time

Cell question

slot 129

c	learned, no gene
–	no rank
$T[t]$	time only

What enters the predictor

$$[\, q_1, \ldots, q_{128}, \; q_{\text{cell}} \,] \in \mathbb{R}^{B \times 129 \times 768}$$

$q_{\text{cell}} = c + T[t]$, with $c \in \mathbb{R}^{768}$. The 19 mask slots are the questions that train $\mathcal{L}_{\text{mask}}$.